$$\frac{\mathrm{d}\mathbf{\Gamma}(\boldsymbol{\gamma}(t))}{\mathrm{d}t} = \mathfrak{L}\,\mathbf{\Gamma}(\boldsymbol{\gamma}(t)) \quad \mathbf{\Gamma}(\boldsymbol{\gamma}(0)) \equiv \mathbf{\Gamma}(\boldsymbol{\gamma}_0) \quad \mathfrak{L} \equiv iL \coloneqq \{\ldots, \mathcal{H}\} \quad \boldsymbol{\gamma} \in X = \mathbb{R}^{6M^{FG}}$$

$$\mathbf{A}(\boldsymbol{\gamma}) = \left\{ A_I(\boldsymbol{\gamma}) \in L^2\left(X, \mathbb{P}\right), \text{ where } I = 1, \ldots, M^{CG} \text{ and } M^{CG} \leq M^{FG} \right\}$$

$$\mathcal{P}_{\mathbf{a}} F(\boldsymbol{\gamma}) = \frac{1}{\Omega(\mathbf{a})} \int_X F(\boldsymbol{\gamma}')\, \delta\left(\mathbf{\Gamma}(\boldsymbol{\gamma}') - \mathbf{a}\right) \rho^{eq}(\boldsymbol{\gamma}') \mathrm{d}\boldsymbol{\gamma}', \qquad \Omega(\mathbf{a}) = \int_X \delta\left(\mathbf{\Gamma}(\boldsymbol{\gamma}') - \mathbf{a}\right) \mathbb{P}(\mathrm{d}\boldsymbol{\gamma}')$$

$$\mathcal{P}_{\mathbf{a}}^{+} = \mathcal{P}_{\mathbf{a}}$$

$$\frac{\partial \mathbf{R}_K}{\partial t} = \frac{\mathbf{P}_K}{M_K}$$

$$\frac{\partial \mathbf{P}_K}{\partial t} = -\omega_{N^A K} \frac{\partial U^{AA}}{\partial \mathbf{R}_K} - \frac{\partial V^{\text{eff}}}{\partial \mathbf{R}_K} - \sum_{J=1}^{M^{CG}} \int_0^t \mathcal{M}_{KJ}(t-s, s)\, M_J^{-1}\, \mathbf{P}_J(t-s)\, \mathrm{d}s + \beta^{-1} \sum_{J=1}^{M^{CG}} \int_0^t \frac{\partial \mathcal{M}_{KJ}(s)}{\partial \mathbf{P}_J}\, \mathrm{d}s + \boldsymbol{\mathcal{F}}_K(t)$$

$$\mathcal{M}_{KJ}\left(\mathbf{\Gamma}, t\right) = \beta\, \left\langle \boldsymbol{\mathcal{F}}_K(0) \otimes \boldsymbol{\mathcal{F}}_J(t) \right\rangle_{\mathbf{\Gamma}}$$

$$\boldsymbol{\mathcal{F}}_K(t) = \mathbf{e}^{t\mathfrak{L}_{\mathcal{Q}}} \mathfrak{L}_{\mathcal{Q}} \mathbf{P}_K$$

$$\mathcal{H}^X(\mathbf{r}, \mathbf{p}) = \frac{1}{2}\, \mathbf{p}^T \left(M^X\right)^{-1} \mathbf{p} + \sum_{i=1}^{M^{FG}-1} U_{i,i+1}^X\left(\mathbf{r}_{i+1} - \mathbf{r}_i\right) + U_{M^{FG},1}^X\left(\mathbf{r}_1 - \mathbf{r}_M^{FG} + r^* M^{FG}\right),$$

$$U_{i,i+1}^X(r) = \epsilon_{i,i+1} \left[\left(\frac{r_{i,i+1}^*}{r} \right)^{12} - 2 \left(\frac{r_{i,i+1}^*}{r} \right)^{6} \right], \quad \epsilon_{i,i+1} = \begin{cases} \sqrt{\epsilon_\alpha \epsilon_\beta} & i = 1,\, 2,\, 4,\, 5,\, \ldots,\, M^{FG}-2,\, M^{FG}-1; \\ \epsilon_\alpha \;\; (\text{or } \epsilon_\beta) & i = 3,\, 6,\, \ldots,\, M^{FG}; \end{cases}$$

$$\widetilde{\Sigma} = \lambda^{-1} \Sigma(\widetilde{\mathbf{R}}), \quad \Sigma_{IJ} = \begin{cases} -\gamma_{I+1,I} & I = J-1 \mod M^{CG}, \\ \gamma_{I+1,I} + \gamma_{I,I-1} & I = J, \\ -\gamma_{I,I-1} & I = J+1 \mod M^{CG}, \end{cases} \qquad \gamma_{I+1,I} \coloneqq \mathbb{E}\left[\sigma_{I+1,I} \otimes \sigma_{I+1,I} \middle| \widetilde{\mathbf{R}} \right].$$